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ORF307: Optimization

Spring 2022 2^{nd} Midterm Exam Solutions

April 15, 11:00am – 12:20pm

Exam Rules

1. Please write out and sign the following pledge on the solution booklet: "I pledge my honor that I have not violated the Honor Code or the rules specified by the instructor during this exam."
2. You are allowed a single sheet of paper, double-sided, hand-written or typed.
3. You cannot communicate with anyone during the exam.
4. All questions should be answered in the solution booklet. Write your name on every page that you use to write the solutions. Start each problem on a separate page. Answering multiple subparts of the same problem on the same page is fine. We will **not** grade anything written on the exam questions; we will only grade what you write in the solution booklet.
5. There are 3 problems worth a total of 100 points.
6. At the end of the exam, please turn in both your solution booklet and the exam questions.

1. *Problem 1 (35 points).*

Consider the problem

$$\begin{array}{ll} \text{minimize} & -2x_1 - x_2 \\ \text{subject to} & 3x_1 + 2x_2 \leq 5 \\ & -x_1 + 3x_2 \geq -4 \\ & x_1, x_2 \geq 0 \end{array}$$

1. (5 points) Convert the problem into standard form and construct a basic feasible solution at which $(x_1, x_2) = (0, 0)$

2. (10 points) Carry out one iteration of the simplex method, with Bland's rule. In particular, report the values for $\bar{c}$, θ, d, and x (including any additional variables you added to make it into standard form) for this iteration. You should explain how you compute each one of these quantities.

3. (5 points) Compute the next point and $\bar{c}$. What would you do if you wanted to complete the simplex method?

4. (7 points) Give the first two values of the optimal dual solution $y^\star$, $(y_1^\star, y_2^\star)$ for the LP. **Hint:** use complementary slackness

5. (8 points) If we change the problem to the one below, where $u = (1, 0)$, without applying simplex again, what is the modified optimal value $p^\star(u)$ and optimal solution $x^\star(u)$? You may assume that the optimal basis is unchanged.

$$\begin{array}{ll} \text{minimize} & -2x_1 - x_2 \\ \text{subject to} & 3x_1 + 2x_2 \leq 5 + u_1 \\ & -x_1 + 3x_2 \geq -4 - u_2 \\ & x_1, x_2 \geq 0. \end{array}$$

Solution

Part 1. Original problem

$$\begin{array}{ll} \text{minimize} & -2x_1 - x_2 \\ \text{subject to} & 3x_1 + 2x_2 \leq 5 \\ & -x_1 + 3x_2 \geq -4 \\ & x_1, x_2 \geq 0 \end{array}$$

In standard form

$$\begin{aligned} \text{minimize} \quad & -2x_1 - x_2 \\ \text{subject to} \quad & 3x_1 + 2x_2 + s_1 = 5 \\ & x_1 - 3x_2 + s_2 = 4 \\ & x_1, x_2, s_1, s_2 \geq 0 \end{aligned}$$

An initial basic feasible solution is $(0, 0, 5, 4)$

Part 2.

$\bar{c} = c - A^T p$ where $A_B^T p = c_B$. Here we have $B = (3, 4)$, $c_B = (0, 0)$ and

$$A_B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

So $p = (0, 0)$ and $\bar{c} = (-2, -1, 0, 0)$. Using Bland's rule, the variable that enters the basis is x_1. The search direction d solves the system $A_B d_B = -A_j$. Here, $A_j = (3, 1)$ so $d = (1, 0, -3, -1)$. We see that $\theta = 5/3$ to force s_1 to leave the basis. The new x is $x = (\frac{5}{3}, 0, 0, \frac{7}{3})$.

Part 3.

$\bar{c} = c - A^T p$ where $A_B^T p = c_B$. Now we have $B = (1, 4)$, $c_B = (-2, 0)$, and

$$A_B = \begin{bmatrix} 3 & 0 \\ 1 & 1 \end{bmatrix}$$

To solve $A_B^T p = c_B$, we solve

$$\begin{aligned} 3p_1 + p_2 &= -2 \\ p_2 &= 0 \end{aligned}$$

and get $p = (-\frac{2}{3}, 0)$, and $\bar{c} = c - A^T p = (0, \frac{1}{3}, \frac{2}{3}, 0)$. Since $\bar{c} \geq 0$, we know that we just found the optimal solution: $x = (\frac{5}{3}, 0, 0, \frac{7}{3})$ and we stop the algorithm.

Part 4.

The dual problem is

$$\begin{aligned} \text{maximize} \quad & -5y_1 - 4y_2 \\ \text{subject to} \quad & 3y_1 + y_2 - y_3 - 2 = 0 \\ & 2y_1 - 3y_2 - y_4 = 0 \\ & y \geq 0 \end{aligned}$$

Note: transform the problem to the inequality form LP, then take the dual. We observe that the second primal constraint is not tight ($x_1 - 3x_2 = \frac{5}{3} < 4$), so by complementary slackness, the second dual variable y_2^* must be 0. Then, to get a dual objective equivalent to the primal objective, $-5y_1^* = -\frac{10}{3}$, so $y_1^* = \frac{2}{3}$. Similarly using complementary slackness and the constraints, we observe that $y_3^* = 0, y_4^* = \frac{1}{3}$. Therefore, by strong duality, $y^* = (\frac{2}{3}, 0, 0, \frac{1}{3})$ must be an optimal dual solution. We only need the first two components, $(\frac{2}{3}, 0)$.

Part 5. Since the shadow prices $= y^* = (\frac{2}{3}, 0)$, and we are given that the optimal basis is unchanged when we perturb the right-hand-side by $u = (1, 0)$, the new optimal value is

$$\begin{aligned} p^*(u) &= p^*(0) - y^{*T}u \\ &= -\frac{10}{3} - \frac{2}{3} \\ &= -4 \end{aligned}$$

The new optimal solution is

$$x_B^*(u) = x_B^* + A_B^{-1}u$$

where

$$A_B = \begin{bmatrix} 3 & 0 \\ 1 & 1 \end{bmatrix}.$$

To solve $A_B q = u$, we solve

$$\begin{aligned} 3q_1 &= 1 \\ q_1 + q_2 &= 0 \end{aligned}$$

so $q_1 = \frac{1}{3}$, $q_2 = -\frac{1}{3}$ Therefore

$$x_B^*(u) = x_B^* + A_B^{-1}u = (\frac{5}{3}, \frac{7}{3}) + (\frac{1}{3}, -\frac{1}{3}) = (2, 2).$$

It follows that $x^*(u) = (2, 0, 0, 2)$.

2. *Problem 2 (35 points).*

Consider the following parametric optimization problem where x is the decision variable and $\theta \in \mathbf{R}$ is a scalar parameter.

$$\begin{array}{ll} \text{minimize} & (c+\theta d)^T x \\ \text{subject to} & Ax = b + \theta f \\ & x \geq 0 \end{array}$$

1. (10 points) Suppose we are given the optimal basis B for $\theta = 0$. Find, (as a function of the problem data and A_B and c_B) the optimal primal solution, $x_B(\theta)$, and the optimal dual solution, $y(\theta)$, locally around $\theta = 0$. Assume that θ is small enough in magnitude such that the optimal basis does not change.

2. (10 points) Suppose that a certain basis B is optimal for $\theta = -10$ and $\theta = 10$. Show that the same basis is optimal for $\theta = 5$.

3. (10 points) Let $b = 0$ and $c = 0$. Suppose that a certain basis B is optimal for $\theta = 1$. For what other values of θ is the same basis optimal?

4. (5 points) Still assume that $b = 0$ and $c = 0$. Show that the optimal value, $p^\star(\theta)$ is quadratic for all θ restricted such that the optimial basis does not change.

Solution

1. By definition of the basis, $x_N(\theta) = 0$. Since B is the optimal basis at $\theta = 0$, $x_B(0) = A_B^{-1}b$ and $y(0) = -A_B^{-T}c_B$. $x_B(0) > 0$ and thus the solution is non-degenerate. Therefore, around $\theta = 0$, the basis B remains optimal. We get $x_B(\theta) = A_B^{-1}(b + \theta f)$ and $y(\theta) = -A_B^{-T}(c_B + \theta d_B)$.

2. It must be true that $x_B(10) \geq 0$ and $x_B(-10) \geq 0$. Since $x_B(5) = \frac{3}{4}x_B(10) + \frac{1}{4}x_B(-10)$, it also holds that $x_B(5) \geq 0$. Since $x_N(\theta) = 0$, primal feasibility holds. To check for optimality we find the reduced costs, $\bar{c}(\theta)$.

$$\bar{c}(\theta) = -A^T A_B^{-T}(c_B + \theta d_B) + c + \theta d = -A^T A_B^{-T} c_B + c + \theta(-A^T A_B^{-T} d_B + d)$$

We see that $\bar{c}(5) = \frac{1}{4}\bar{c}(-10) + \frac{3}{4}\bar{c}(10)$. Since $\bar{c}(-10) \geq 0$ and $\bar{c}(10) \geq 0$, we have that $\bar{c}(5) \geq 0$ and we have primal optimality. The zero duality gap also holds since we have a basic feasible solution (zero duality gap always holds for every iteration of the simplex method).

3. Here, we can simplify the formulas given the optimal basis as $x_B(\theta) = \theta A_B^{-1} f$ and $y(\theta) = -\theta A_B^{-T} d_B$. The reduced cost vector becomes $\bar{c}(\theta) = \theta(-A^T A_B^{-T} d_B + d)$. But since B is the optimal basis for $\theta = 1$, we see that $x_B(1) = A_B^{-1} f \geq 0$. And $\bar{c}(1) = -A^T A_B^{-T} d_B + d \geq 0$. We can write our formulas for general θ in terms of $x_B(1)$ and $\bar{c}(1)$. We get $x_B(\theta) = \theta x_B(1)$ and $\bar{c}(\theta) = \theta\bar{c}(1)$. Thus this basis is optimal if and only if $\theta \geq 0$.

4. $p^\star(\theta) = \theta^2 d^T A_B^{-1} f$ for $\theta \geq 0$. This is seen from using the optimal x in the previous part and plugging it into the objective. We also see that the strong duality holds when doing the same for checking the objective of the dual.

3. *Problem 3 (30 points).*

For a given matrix A, show that exactly one of the following alternatives must hold.

(a) $x_1 > 0$ for every vector x such that $Ax \geq 0, x \geq 0$. Note that x_1 is the first element of x.

(b) There exists some vector y such that $A^T y \leq 0$, $y \geq 0$, and $A_1^T y < 0$. Note that A_1 is the first column of A.

Hint: For the primal problem relating to alternative (a), do not put all of the conditions as constraints in the feasible region.

Solution

¡!–First, recognize that we can replace (b) with the following condition:

(c) There exists some vector y such that $A^T y \leq 0, y \geq 0$, and $A_1^T y \leq -1$.

To see this, recognize that if (c) holds, then (b) clearly holds.

On the other hand, if (b) holds, then let $\varepsilon = A_1^T y < 0$. Then, the vector

$$z = -y/\varepsilon$$

satisfies $A^T z \leq 0$, $z \geq 0$, and $A_1^T z \leq -1$. So, by this scaling argument, (b) implies (c).

Since (b) and (c) are equivalent, we can instead show that (a) and (c) are strong alternatives. –¿

Let e_1 be the vector with a 1 as the first entry and 0 everywhere else. Construct the following primal linear program:

$$\begin{array}{ll} \text{minimize} & -e_1^T x \\ \text{subject to} & Ax \geq 0 \\ & x \geq 0 \end{array}$$

and the corresponding dual:

$$\begin{array}{ll} \text{maximize} & 0 \\ \text{subject to} & A^T y \leq -e_1 \\ & y \geq 0 \end{array}$$

Recognize that $x = 0$ is feasible for the primal, so by strong duality, the optimal solutions match, i.e. $p^\star = d^\star$. Since the dual has objective 0, there are two cases:

Case 1) If there exists y that satisfies (b), then the optimal solution of the dual is 0. This is because if some y satisfies $A_1^T y < 0$, then we can define $z = -y/(A_1^T y)$ such that $A^T z \leq -e_1, z \geq 0$. This means that (b) corresponds to the dual being feasible. By strong duality, $p^\star = d^\star = 0$. In order for the primal to match, it must be bounded, which happens only if $x_1 = 0$, which violates condition (a).

Case 2) If there exists no y that satisfies (b), then then $p^\star = d^\star = -\infty$ since the dual is a maximization that is infeasible. In this case, the primal must be unbounded, which happens only if $x_1 > 0, Ax \geq 0$, and $x \geq 0$, because for any solution x, αx is a solution for $\alpha > 0$, so condition (a) holds.